

Finding Inverses with Domain Restrictions

Answer Key

Precalculus

Quick Answers

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|--|---|
| 1. 500 min | 6. (a) time = 4 s, — |
| 2. x , — | 7. -15 degrees |
| 3. (a) domain of the inverse = $(0, \infty)$, (b) radius = 8 cm | 8. (a) domain of the inverse = $[-4, \infty)$, (b) inverse value = 8 |
| 4. 25 kg | 9. x , — |
| 5. x , — | 10. (a) time = 2 s, — |
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What is verified

12 of 17 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 2, 5, 6, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-BF.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

8. *If they got -2 : took the minus square root, which is the branch the restriction $x \geq 3$ removes*

1. $C(m) = 0.05m + 20$ dollars for m minutes. A bill of 45 dollars — find m .

Solution: Swap and solve: from $C = 0.05m + 20$ we get $m = \frac{C - 20}{0.05}$, so $C^{-1}(C) = \frac{C - 20}{0.05}$. At $C = 45$: $\frac{45 - 20}{0.05} = \frac{25}{0.05} = 500$. $m = 500$ min

2. $f(x) = 4x + 9$, $g(x) = \frac{x - 9}{4}$. Show $(f \circ g)(x) = x$.

Solution: $f\left(\frac{x - 9}{4}\right) = 4 \cdot \frac{x - 9}{4} + 9 = (x - 9) + 9$. The 4 and the division by 4 cancel, then the $+9$ undoes the -9 . $(f \circ g)(x) = x$

Grading the shown work (*instructor-judged*): the value alone is not the answer here. Look for the substituted line and the cancellation, ending at x . A bare “ $= x$ ” with no intermediate step, or a check at a single numerical input, does not establish the identity.

3. $A(r) = \pi r^2$ for $r > 0$ (radius in cm, area in square cm). (a) Domain of A^{-1} . (b) Radius when $A = 64\pi$.

Solution (a): The domain of the inverse is the range of A . As r runs over $r > 0$, πr^2 takes every positive value and no others, so the domain of A^{-1} is $A > 0$

Solution (b): Solve $A = \pi r^2$ for r : $r^2 = \frac{A}{\pi}$, and the restriction $r > 0$ picks the positive root, so $A^{-1}(A) = \sqrt{A/\pi}$. At $A = 64\pi$: $\sqrt{64\pi/\pi} = \sqrt{64} = 8$. $r = 8$ cm

4. $L(w) = 12 + 0.4w$ cm for a load of w kg. Find w when $L = 22$ cm.

Solution: From $L = 12 + 0.4w$, $w = \frac{L-12}{0.4}$. At $L = 22$: $\frac{22-12}{0.4} = \frac{10}{0.4} = 25$.

$$w = 25 \text{ kg}$$

5. $f(x) = \frac{2x+1}{x-3}$, $g(x) = \frac{3x+1}{x-2}$. Show $(f \circ g)(x) = x$.

Solution: Substitute and clear the compound fraction by multiplying numerator and denominator by $x-2$:

$$f(g(x)) = \frac{2\left(\frac{3x+1}{x-2}\right) + 1}{\left(\frac{3x+1}{x-2}\right) - 3} = \frac{(6x+2) + (x-2)}{(3x+1) - 3(x-2)} = \frac{7x}{7}$$

$$(f \circ g)(x) = x$$

Grading the shown work (*instructor-judged*): look for the compound fraction actually being cleared — the substituted line, then the multiplication by $x-2$ top and bottom, then $7x/7$. A bare “ $= x$ ” or a single numerical check earns no credit.

6. $d(t) = 5t^2$ metres for $t \geq 0$ seconds. (a) Time to fall 80 metres. (b) Why the restriction matters.

Solution (a): From $d = 5t^2$, $t^2 = \frac{d}{5}$, and $t \geq 0$ selects the positive root: $d^{-1}(d) = \sqrt{d/5}$. At $d = 80$: $\sqrt{16} = 4$. $t = 4$ s

Solution (b): *Instructor-judged*. The rule squares the time, so on all real numbers t and $-t$ give the same distance and the assignment distance-to-time would not be a function at all. Restricting to $t \geq 0$ leaves exactly one time for each distance — the rule becomes one-to-one — and that is what lets the square root be taken with a single sign. A student who only says “negative time is impossible” has given the physical reason, not the mathematical one; credit that answer fully only if the one-to-one point is also made.

7. $F(c) = \frac{9}{5}c + 32$. Find c when $F = 5$.

Solution: Undo the operations in reverse order — subtract 32, then multiply by $\frac{5}{9}$:

$$c = \frac{5}{9}(F - 32) = \frac{5}{9}(5 - 32) = \frac{5}{9}(-27)$$

$$c = -15 \text{ degrees Celsius}$$

8. $f(x) = x^2 - 6x + 5$ on $x \geq 3$. **(a)** Domain of f^{-1} . **(b)** $f^{-1}(21)$.

Solution (a): Completing the square gives $f(x) = (x - 3)^2 - 4$, so the vertex sits at $x = 3$ — exactly where the restriction starts. On $x \geq 3$ the outputs run from -4 upward, so the domain of f^{-1} is

$$x \geq -4$$

Solution (b): Solve $y = (x - 3)^2 - 4$ for x : $(x - 3)^2 = y + 4$, and $x \geq 3$ forces the plus sign, so $f^{-1}(x) = 3 + \sqrt{x + 4}$. Then $f^{-1}(21) = 3 + \sqrt{25} = 3 + 5 = 8$.

$$f^{-1}(21) = 8$$

9. $f(x) = \frac{1}{x-4}$, $g(x) = \frac{1}{x} + 4$. Show $(f \circ g)(x) = x$.

Solution: The $+4$ inside g is exactly what the -4 inside f removes:

$$f(g(x)) = \frac{1}{\left(\frac{1}{x} + 4\right) - 4} = \frac{1}{\frac{1}{x}}$$

and the reciprocal of $\frac{1}{x}$ is x .

$$(f \circ g)(x) = x$$

Grading the shown work (*instructor-judged*): look for the $+4$ and -4 cancelling inside the denominator, then the reciprocal step. A bare “ $= x$ ” or a single numerical check earns no credit.

10. $h(t) = 45 - 5t^2$ metres for $0 \leq t \leq 3$ seconds. **(a)** Time at height 25 metres. **(b)** Why the restriction is needed.

Solution (a): From $h = 45 - 5t^2$: $5t^2 = 45 - h$, so $t^2 = \frac{45 - h}{5}$ and the restriction $t \geq 0$ gives $h^{-1}(h) = \sqrt{\frac{45 - h}{5}}$. At $h = 25$: $\sqrt{\frac{20}{5}} = \sqrt{4} = 2$.

$$t = 2 \text{ s}$$

Solution (b): *Instructor-judged*. Because the rule uses t^2 , every height below 45 metres is produced by two times, t and $-t$, so without a restriction no single inverse rule could decide which to report. Cutting the domain to $0 \leq t \leq 3$ leaves exactly one time per height, which is what makes the model one-to-one and invertible. The upper end matters too: at $t = 3$ the height reaches 0, and past that the model describes nothing. An answer that objects only to negative time earns partial credit.